

Debt issuance and option repricing

H23 | Massive track | October 4, 2026

1. Summary

Does a standard debt-issuance disclosure identify an option-overlay opportunity beyond prior prices and option premiums? Both covered-call and protective-put branches are fixed before implementation. Verified pilot development output; fresh OOS unavailable. Verdict: UNSUPPORTED_FOR_CONFIRMATORY_CLAIM.

2. Economic Hypothesis

Financing terms can change assessments of funding access and leverage. If funding uncertainty is resolved before option prices fully adjust, selling a call against funded shares may earn an unusual net increment. If new leverage or financing pressure instead leaves downside underpriced, purchased protection may reduce losses within a fixed premium budget. Neither implication follows automatically from the label.

The proposed persistence is different speeds of interpreting financing details. Call buyers seek upside exposure; put sellers accept contingent downside for premium. These participant descriptions are proposed mechanisms, not observed order-flow findings. Ordinary option carry, stock exposure, already priced financing information, late entry or friction can explain or remove the proposed increment. A completion or receipt-of-funds phrase is not required for primary eligibility.

Complete economic registration b4159e2917248d588072ef5d9ae55bd88c3f8063 preceded the new implementation. Both branches are retained; no favorable issuer or option payoff chooses the primary rule.

3. Data & Universe

The static sponsor universe contains 100 companies. Financial inputs are Massive API data for January 1, 2022 through December 31, 2025: disclosures, historical standard contracts, nominal stock and option bars, available quotes, splits and dividends. H21 served 526 debt-issuance filings across 87 issuers. These are raw source counts, not usable matched pairs or executed trades.

All 2022-2025 is exposed development. Earlier 2026 research is exposed and cannot provide fresh confirmation. Historical membership, service-release clocks and prior public-announcement timing remain limitations. The organizer's sealed interval stays excluded.

4. Methodology

Signal: debt_issuance. Use every served primary label with no completion, receipt or excerpt-word gate. CIK/accession identity preserves bundles; the lexicographically first mapped share class is chosen before outcomes. Same-issuer decisions on one date are consolidated.

Assume availability at the session aligned to filing plus one calendar day, then add 1 session(s). Decide and select contracts at that close; enter at the next session close. Target expiry nearest 120 calendar days within 90-180 DTE and strikes near 5% OTM within one percentage point. Require verified 100-share deliverables. Buy 100 actual shares with one short call or one long put; exit ten sessions after entry.

Match one same-issuer ordinary date 84-252 earlier sessions using strictly prior momentum, RMS volatility, volume and SPY conditions. Ordinary dates have no served bundle in their preceding five sessions. Compare complete event/ordinary strategy and stock-only quartets, with matched DTE, moneyness and premium calipers. H25 additionally uses earlier debt-only and underwriting-only controls; absent component matches prevent interaction claims.

Per-side research friction is stock 6 basis points; options 5% of premium plus \$0.65 per contract. These are assumptions, not measured spreads. Fresh synchronized quote checks remain separate. Charge all premiums, dividends and modeled assignment/corporate-action cash flows. Marked trade bars do not establish executable closing fills.

All eight horizons (1, 2, 3, 5, 10, 21, 42, 63 sessions) and observable expiry are retained. Eleven variants per study cover costs, delay, maturity, moneyness, assignment and size. The shared S01-S12 specification controls training-only fitting, attribution, matching and exclusions; the 19-effect family uses 9,999 joint issuer/calendar draws at 63 and 126 sessions after synthetic-null calibration. Both lengths must pass. The focused 199 timing placebos do not replace H21's unfinished full-atlas placebo family.

Measured calibration gate

Block sessions	False rejections / datasets	Allowed	Status
63	45 / 100	10	fail
126	72 / 100	10	fail

This is synthetic estimator calibration, not market-performance evidence. Status: fail. Failure blocks inferential significance and practical-bound claims. Descriptive pilot estimates remain visible; no seed retry or threshold change repairs this result.

5. Results

Verified pilot development output; fresh OOS unavailable. Performance below is sourced only from verified returned tables when supplied. Unrun, unsupported and unavailable quantities remain explicit. A positive funded stock return does not identify an event-timing contribution.

Primary event portfolio	Covered call	Protective put
Annualized return	0.016%	Unavailable
Annualized volatility	0.129%	Unavailable
Sharpe	0.126	Unavailable
Maximum drawdown	-0.243%	Unavailable
Turnover	22.800%	Unavailable
Sessions	1003.000	1003.000
Stock-only annualized return	0.075%	-0.021%

Observed tasks: 2; planned: 11. Coverage: priced_subset_conditional; full primary-label cohort preserved in prepared/coverage.csv. Calibration: fail; block inference: suppressed_fail; placebos: pending.

Primary effect / comparator	Estimate (pp)	Pairs / issuers
Call overlay / ordinary	2.250	1 / 1
Call strategy / ordinary	-1.142	1 / 1
Put hedge / ordinary	-2.245	2 / 1
Put downside / ordinary	-0.041	2 / 1
Put return drag / ordinary	-0.041	2 / 1

Effect estimates are percentage-point differences over the ten-session horizon, not annualized alpha. Evidence statuses: insufficient support. The registered overall floor is 30 pairs across eight issuers. This coverage-only pilot cannot replace the full cohort.

Fresh OOS: unavailable. All historical output in this round is development evidence. A registered failure is retained as rejected; missing support or required evidence is inconclusive. No supported incremental-signal verdict is permitted from exposed history alone.

Results continued: response, curves and robustness

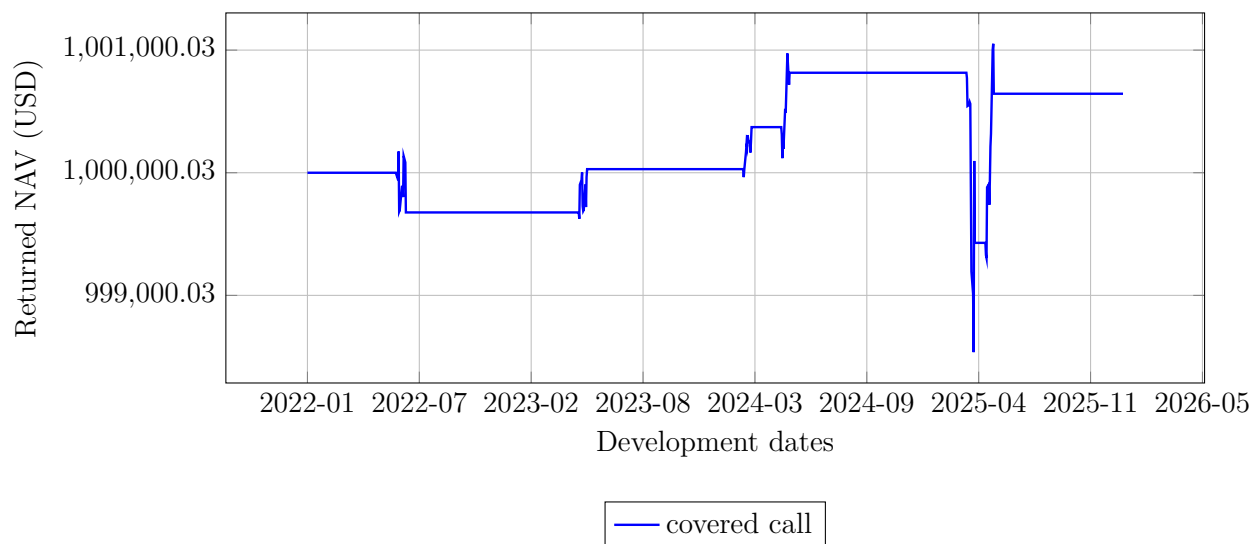


Figure: returned primary event-portfolio NAV, including the run's cash and cost convention. The curve is a display of retained scheduled output; it is not an OOS result, ablation test or independent replication. Baseline tables retain the comparable stock-only and ordinary-date evidence.

Unavailable curves: protective put - complete matching event-portfolio performance unavailable.

Fixed advancement thresholds: call overlay lower bound >0.10 percentage points and strategy increment >0 ; put hedge benefit >0 , downside reduction >0.25 points and return drag >-0.50 points. Overall support is 30 pairs/eight issuers; annual support ten pairs/five issuers. H25 additionally requires positive contrasts against both components. All numbers must come from the returned comparison tables.

Horizon and robustness results remain pending unless returned. Report every registered horizon, failed mark, expiry boundary, base/doubled cost, extra delay, 30/60-day maturity, 3%/10% OTM, assignment and ten-contract variant. These are diagnostics; they do not authorize a favorable variant to replace the primary.

Prior discovery: financing examples improved full-feature-group absolute-return prediction in each evaluation year, but all displayed adjusted p-values were 1 at both block lengths. Exposure commit 1d531db17acc3cd6f0df6cebdb2ff503b0d40b95 records that selection. H21's 350 cross-year-positive groups overlap; none passed the complete declared screen.

6. Risk Management

Each study/shape starts with \$1 million, zero cash yield, no leverage and one 100-share/one-contract unit. Name exposure is capped at 5% of NAV; funded-share gross/net and sector caps are 20%. Unknown sectors share one conservative bucket. No overlapping position in an issuer is opened. A 5% drawdown requests next-feasible liquidation and a 21-session entry cooldown. Gaps and missing exits can defeat timely liquidation.

Covered calls retain equity downside and cap upside. Protective puts cost premium and depend on valid contract/action/settlement accounting. Modeled early assignment is an assumption, with a conservative sensitivity. Unresolved actions or long mark gaps suppress reliable full-period metrics; they are not filled from future prices.

7. Liquidity & Capacity

Eligibility uses only completed pre-decision volume history. Bounded quote checks require fresh, synchronized, positive sized quotes with registered spread limits. Per-side fees and impact stress are separate from observed bid/ask. Daily volume and snapshot size do not establish intraday depth. Ten-contract stress remains a scenario; dollar capacity cannot be claimed before a supported net benefit and measured liquidity exist.

8. Limitations & Next Steps

Standard labels can be heterogeneous, and same-filing labels may describe different economic transactions. Matching is observational rather than causal. Static membership, historical label delivery, announcement timing and approximate execution remain limitations. Earlier H21/H22 selection, failed inference and inconclusive outcomes are preserved. Complete option coverage, calibrated inference, placebos and genuinely unseen confirmation are required before advancement.

PC work creates code, downloads, immutable packages and paper sources. Actual matching, backtests and inference use HiPerGator allocations; returned archives, member hashes and scheduler states must verify. Overleaf compiles and exports this standalone source. The note is a pending research draft until those evidence and compilation checks complete; it is not a competition-submission receipt.

References and reproduction record

Gator Quant Hacks, Systematic Trading and Massive challenge, retained October 3 review.

<https://www.gqhacks.com/tracks/systematic-trading/massive>

Massive, disclosure taxonomy and supporting-excerpt methodology. Historical delivery clocks remain unverified.

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Massive, historical contract reference; historical as-of selection and deliverables.

<https://massive.com/docs/rest/options/contracts/all-contracts>

Massive, option daily aggregates; access and coverage remain source-specific.

<https://massive.com/docs/rest/options/aggregates/custom-bars>

Massive, historical option quotes; synchronized quotes and complete pagination require separate verification.

<https://massive.com/docs/rest/options/quotes>

The Options Playbook, covered-call construction and continuing equity downside.

<https://www.optionsplaybook.com/option-strategies/covered-call>

The Options Playbook, protective-put construction and premium costs.

<https://www.optionsplaybook.com/option-strategies/protective-put>

Romano and Wolf, dependence-aware resampling and multiple-test methodology. Citation does not validate implementation.

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Local authoritative specification: docs/massive/financing-round2-plan.md and config/financing-round2.json, with this folder's Hypothesis.md and Research Log.md. The Webull/Backtrader starter and Massive starter supply attributed infrastructure/shape references; H22 funded-share accounting is reuse context, not evidence for these hypotheses.

Paper reproduction: scripts/build_financing_paper_sources.py without `-run` creates the registered pending draft. With `-run` it requires the actual verified import, returned member hashes and completed scheduler observation before displaying returned metrics. Upload the generated single-file main.tex ZIP to Overleaf; no key, raw provider object or environment is included.

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Credits: organizer source material, Massive market data, documented option shapes and cited methods informed the study. OpenAI Codex assisted with research/code/documentation and LaTeX preparation. The team is responsible for the code and claims. References are outside the main five-page budget; decisive evidence belongs in the main text. The starter's shorter-write-up instruction and reviewed five-page guidance remain a documented source conflict.